MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

OVERALL SEVERITY: CRITICAL

Case ID	`case01_baseline_vs_case02_t1055_5_attack`
Generated	`2026-04-18 19:45:04 UTC`
Baseline capture	`case01_baseline`
Attack capture	`case02_t1055_5_attack`
Baseline path	`/MFF/cases/case01_baseline`
Attack path	`/MFF/cases/case02_t1055_5_attack`

1. EXECUTIVE SUMMARY

Metric	Value	Metric	Value
New Processes (Attack Only)	13	Critical Risk Processes	2
Gone Processes (Baseline Only)	9	ATT&CK Techniques Observed	2
New Network Connections	131	ATT&CK Tactics Observed	1
Flagged Connections	3	IOCs Extracted	40

ATT&CK; Tactics Observed:

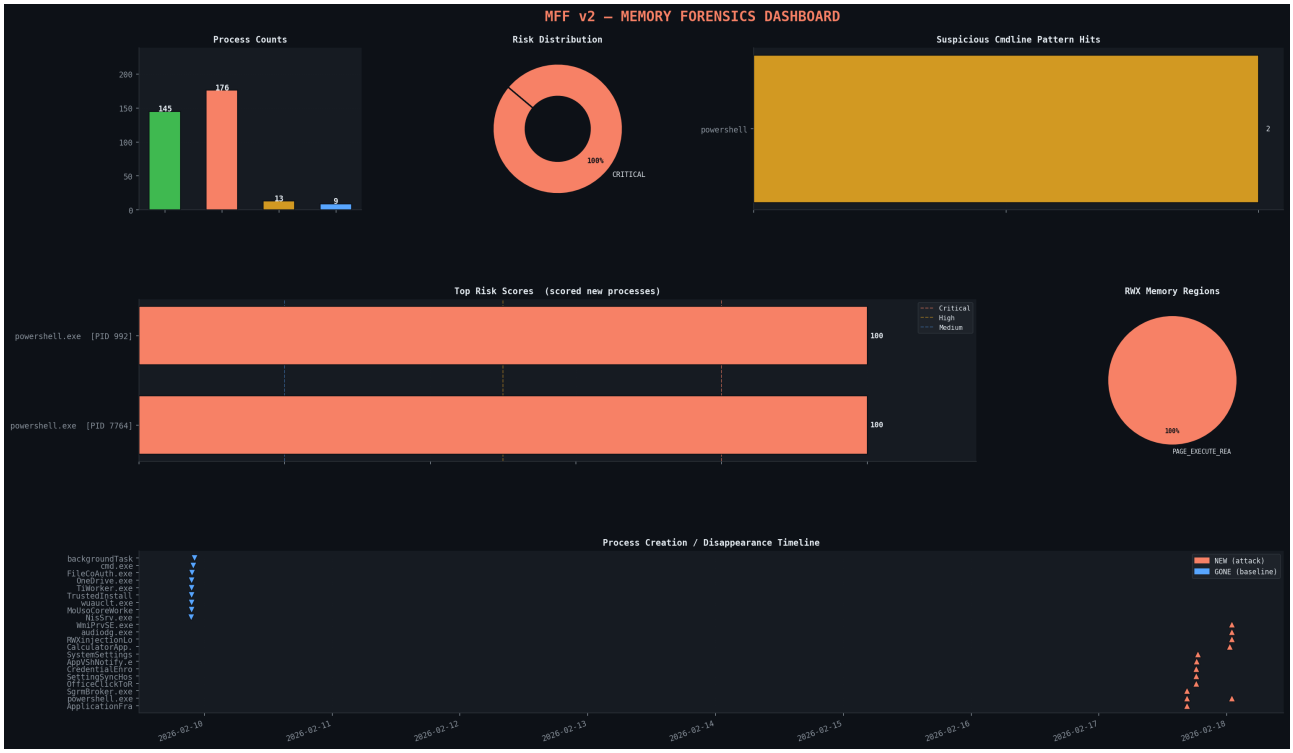
Defense Evasion

IOC Summary:

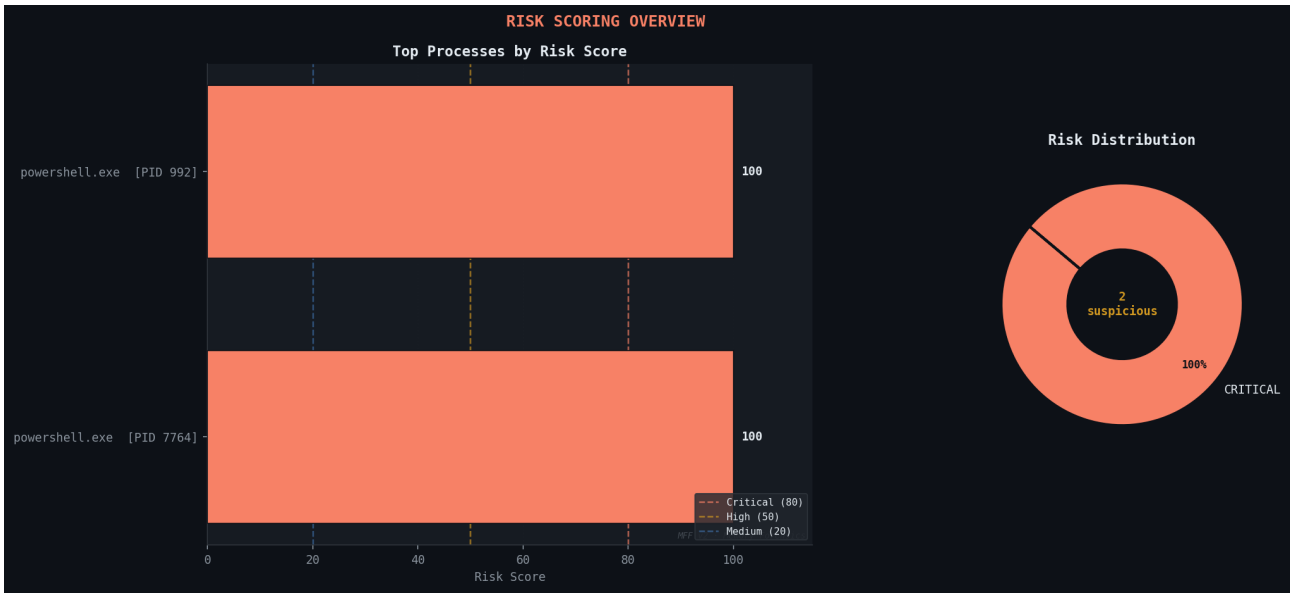
Type	Indicators (top 8)
IPv4	0.0.0.0 · 150.171.28.11

2. VISUALIZATIONS

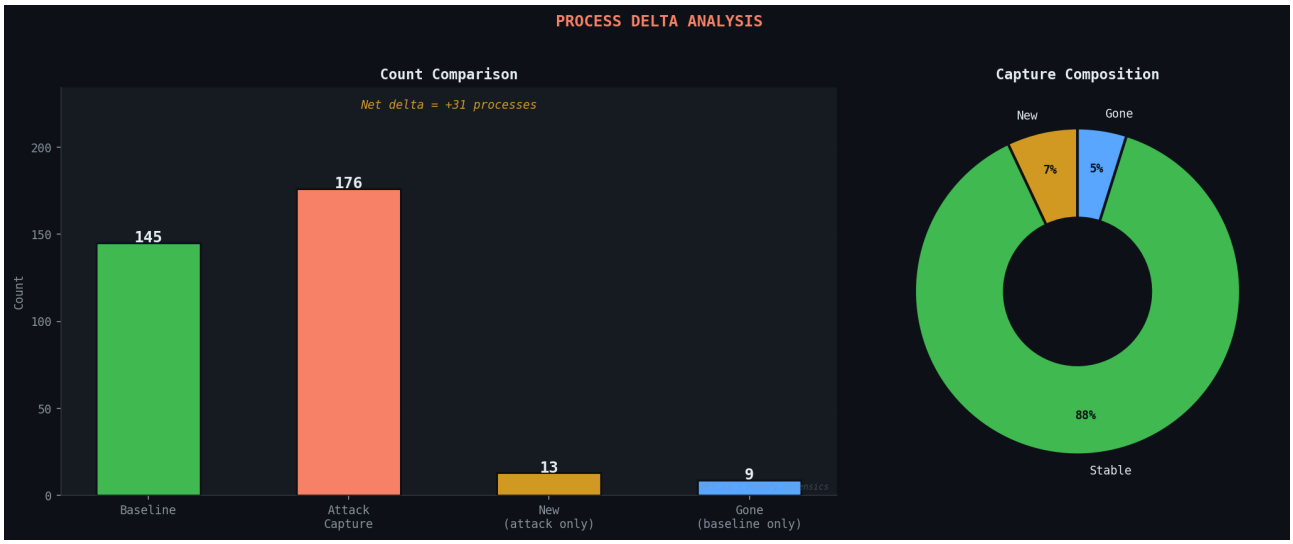
2.1 Full Forensics Dashboard



2.2 Risk Scoring Overview

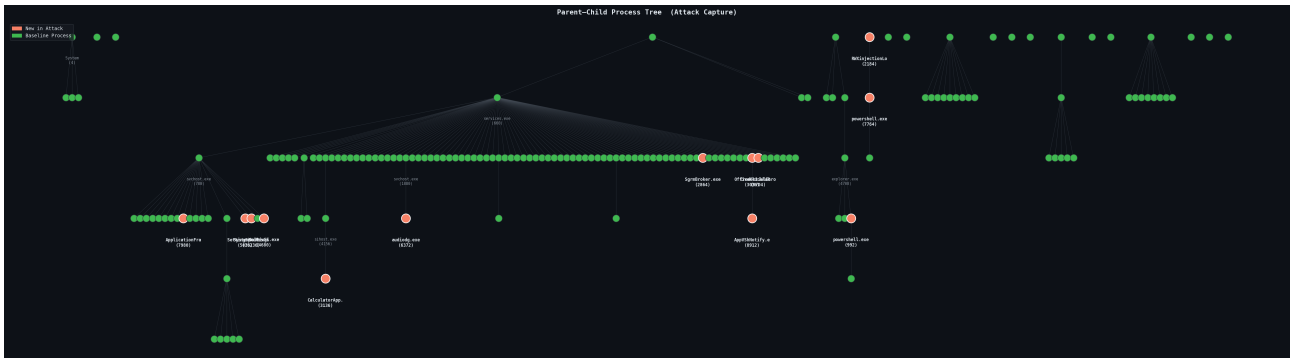


2.3 Process Delta Analysis

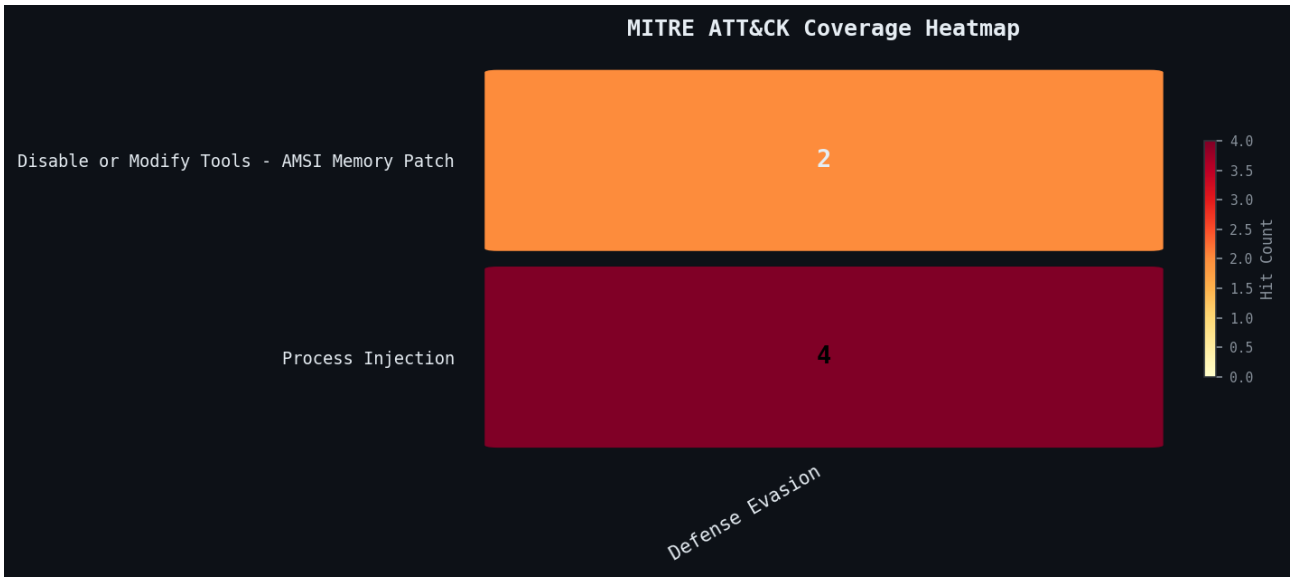


2.4 Process Timeline

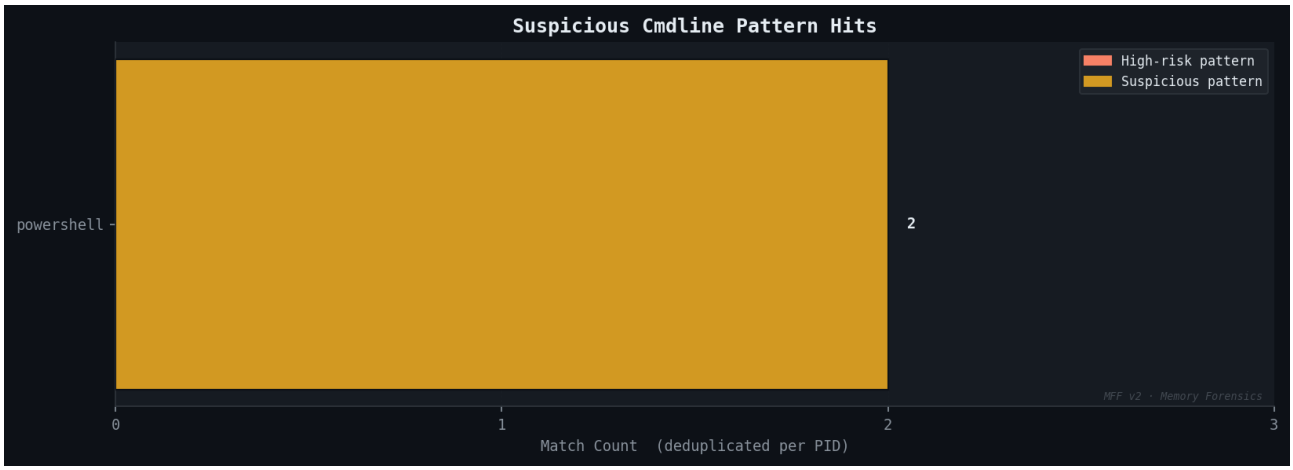
2.5 Parent-Child Process Tree



2.6 MITRE ATT&CK; Heatmap



2.7 Suspicious Cmdline Patterns



2.8 Malfind Memory Protections

3. RISK SCORE TABLE

Process	PID	Score	Level	Indicators		
powershell.exe	992	100	CRITICAL	T1562.001	3 src	High
powershell.exe	7764	100	CRITICAL	T1562.001	3 src	High

4. NEW PROCESSES (Attack Only)

ImageFileName	PID	PPID	CreateTime
ApplicationFra	7980	780	2026-02-17 16:30:03.000000 UTC
powershell.exe	992	4708	2026-02-17 16:30:29.000000 UTC
SgrmBroker.exe	2864	660	2026-02-17 16:30:58.000000 UTC
OfficeClickToR	3016	660	2026-02-17 18:12:01.000000 UTC
SettingSyncHos	5076	780	2026-02-17 18:14:10.000000 UTC
CredentialEnro	9704	660	2026-02-17 18:14:21.000000 UTC
AppVShNotify.e	8912	3016	2026-02-17 18:18:11.000000 UTC
SystemSettings	10236	780	2026-02-17 18:32:00.000000 UTC
CalculatorApp.	3136	4156	2026-02-18 00:31:14.000000 UTC
RWXinjectionLo	2184	9076	2026-02-18 00:55:10.000000 UTC
powershell.exe	7764	2184	2026-02-18 00:55:10.000000 UTC
audiodg.exe	6372	1880	2026-02-18 00:55:11.000000 UTC
WmiPrvSE.exe	4680	780	2026-02-18 00:55:16.000000 UTC

5. MITRE ATT&CK; TECHNIQUE MATCHES

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	992	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	7764	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	992	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	7764	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	992	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	7764	powershell.exe

6. SUSPICIOUS CMDLINE FINDINGS

ImageFileName	PID	MatchedPattern	Args
powershell.exe	992	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
powershell.exe	7764	powershell	\\??\\C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -Command "Add-Type -AssemblyName PresentationFramew

7. MALFIND — MEMORY INJECTION ARTEFACTS

Process	PID	Protection
powershell.exe	992	PAGE_EXECUTE_READWRITE
powershell.exe	992	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE
powershell.exe	7764	PAGE_EXECUTE_READWRITE

7b. DLL ANALYSIS — T1574 / T1562 FINDINGS

2 DLL finding(s) — T1574 DLL Search Order Hijacking + T1562 AMSI Bypass evidence.

PID	Process	DLL	HijackType	Technique	RiskScore	Tactic
992	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
7764	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=992)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=992) loads amsi.dll AND has 2 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=7764)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=7764) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

8. NETWORK ARTEFACTS

8a. Flagged Connections (suspicious ports)

TreeDepth	Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created	ImageFileName	SuspiciousPort	PortMeaning
0	0xdb84c157223 0	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	900	svchost.exe	2026-02-17 15:28:51.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xdb84c157223 0	TCPv6	::	135	::	0	LISTENING	900	svchost.exe	2026-02-17 15:28:51.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xdb84c1572e90	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	900	svchost.exe	2026-02-17 15:28:51.000000 UTC	svchost.exe	135	DCOM/RPC

9. IOC DETAIL TABLE

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\svchost.exe	cmdline	46
FilePath	C:\\Windows\\System32\\svchost.exe	cmdline	29
FilePath	C:\\Program	cmdline	9
FilePath	C:\\Windows\\System32\\RuntimeBroker.exe	cmdline	6
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView	cmdline	4
FilePath	Microsoft\\EdgeWebView\\Application\\144.0.3719.115\\msedgewebview2.exe	cmdline	4
FilePath	??\\C:\\Windows\\system32\\conhost.exe	cmdline	2
FilePath	C:\\ProgramData\\Microsoft\\Windows	cmdline	2
FilePath	C:\\Windows\\system32\\DllHost.exe	cmdline	2
FilePath	Microsoft\\Edge\\Application\\msedge.exe	cmdline	2
FilePath	Windows	cmdline	2
FilePath	system32\\csrss.exe	cmdline	2
FilePath	??\\C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Windows\\Explorer.EXE	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxService.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxTray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	1
FilePath	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\spoolsv.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe	cmdline	1
FilePath	C:\\Windows\\system32\\AUDIODG.EXE	cmdline	1
FilePath	C:\\Windows\\system32\\ApplicationFrameHost.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SearchIndexer.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SecurityHealthService.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SgrmBroker.exe	cmdline	1
FilePath	C:\\Windows\\system32\\lsass.exe	cmdline	1

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\services.exe	cmdline	1
FilePath	C:\\Windows\\system32\\wbem\\wmiprvse.exe	cmdline	1

10. METHODOLOGY

Acquisition: VBoxManage debugvm dumpvmcore — cold hypervisor-level acquisition. No kernel module or guest agent. Forensically sound. SHA256 + MD5 hash recorded on host immediately post-acquisition.

Volatility 3: Memory parsed using Volatility 3: pslist, pstree, cmdline, dlllist, malfind, netscan, threads. Exported as CSV (engine) and JSONL (audit trail).

Process diff: Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs are reassigned on every reboot and produce false-positive churn exceeding 90% of all processes.

Risk scoring: Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline pattern +40 · Known-malicious process name +50 · Suspicious DLL load (T1574) +30 · Anomaly name bonus +15. Thresholds: CRITICAL ≥80 HIGH ≥50 MEDIUM ≥20 LOW <20.

Malfind filter: Known-clean JIT processes excluded (browsers, Defender, .NET, OneDrive). Self-match false positives suppressed (e.g. lsass.exe matching itself). Only genuinely suspicious PAGE_EXECUTE_READWRITE regions retained.

DLL analysis: windows.dlllist CSV analysed for T1574 DLL Search Order Hijacking. Protected DLLs (amsi.dll, version.dll, cryptbase.dll etc.) checked for loads from non-System32 paths. Findings merged into ATT&CK; tagged output.

ATT&CK; mapping: 30+ rule-based signatures across 8 MITRE ATT&CK; tactics. Keyword matching on cmdline args, process names, DLL paths, and network artefacts. T1574-specific patterns include: copy-item, amsi.dll, invoke-atomictest.

IOC extraction: Regex-based extraction: IPv4 (excluding RFC1918 private ranges), domains, URLs, MD5/SHA1/SHA256 hashes, Windows file paths.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project — University of Roehampton 2026. Tool chain: VBoxManage → Volatility 3 → MFF v2 → HTML + PDF reports. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, export_alert, report_generator.

Risk scoring: Transparent additive model — no ML. Each evidence category contributes once. Weights calibrated to specificity (ACPO P2). CRITICAL>=80, HIGH>=55, MEDIUM>=25.

Standards: ACPO Good Practice Guide for Digital Evidence (Principles 1-4). NIST SP 800-86 Integrating Forensic Techniques into Incident Response. ISO/IEC 27037:2012 Guidelines for Identification, Collection and Preservation. College of Policing Digital Forensics Guidance. CPS Disclosure Manual.